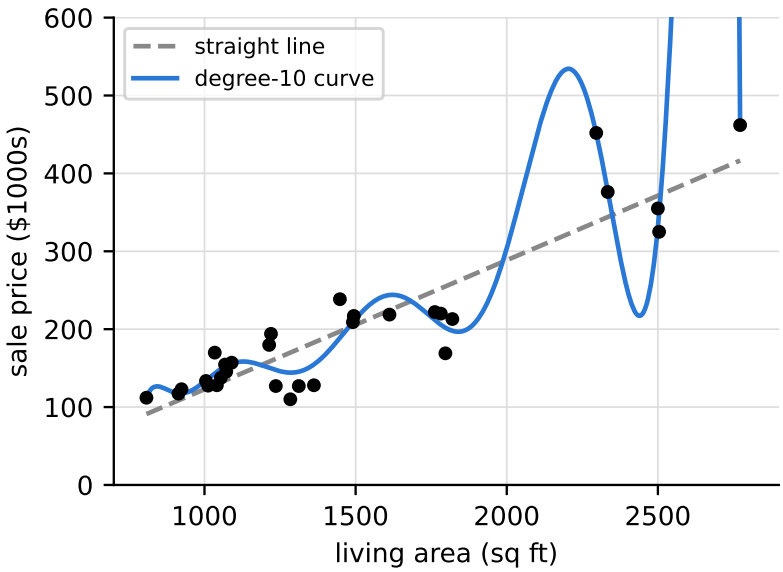
DATA 202 — Ridge and Lasso · Week 10 Wednesday

Sale price vs. living area for houses in Ames, Iowa. Your professor will tell you when to do each part.

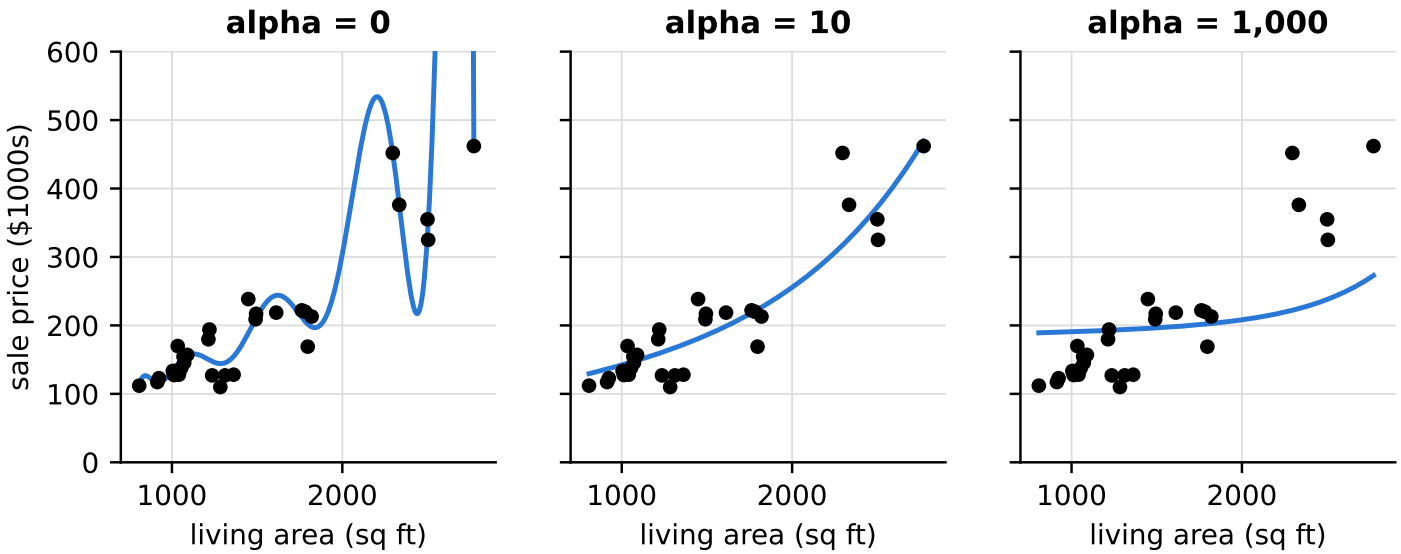
1 · Thirty houses



Thirty sales, a straight line and a degree-10 curve.

1. **Predict:** which model has the smaller RMSE on these 30 houses? _____ On *other* houses of the same sizes? _____
2. **Circle** two places where the curve follows a single house instead of the trend.
3. Which model is overfitting? _____

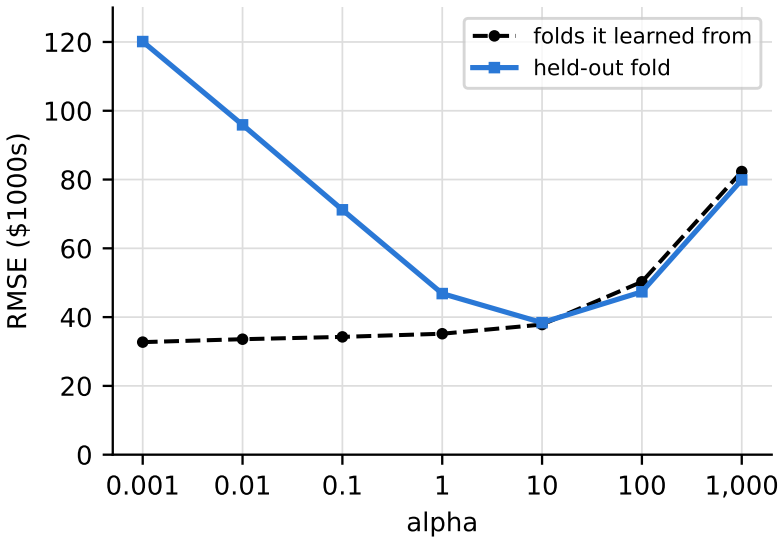
2 · Three alphas



Ridge's curve on the same 30 houses, for three values of alpha.

1. **Write** on each panel: *overfitting*, *about right* or *underfitting*.
2. **Pair talk:** with 30 different houses, which curve would change the most? Which would barely move? Is barely moving a good thing?

3 · Choose alpha



Ridge's typical miss in 5-fold cross-validation on the 30 houses.

1. **Circle** the best alpha.
2. **Label** the overfitting side and the underfitting side.
3. Here a *larger* alpha means a _____ model. For trees, a *larger* `max_depth` meant a _____ model.

4 · What Lasso keeps

Back to all 2,193 training houses and Monday's 47 columns. Monday's line, with no penalty: CV RMSE **$27,876**.

alpha	columns kept (of 47)	CV RMSE
10	44	$27,857
100	32	$28,043
300	25	$28,563
1,000	15	$29,653
3,000	13	$31,310
10,000	9	$35,343
30,000	4	$51,003

1. Find the **simplest** model within $1,000 of Monday's. alpha = _____, with _____ columns.
2. **Mark** the row that is underfitting.
3. **Pair talk:** Lasso dropped `second_floor_area` but kept `living_area`. Why might that cost almost nothing?